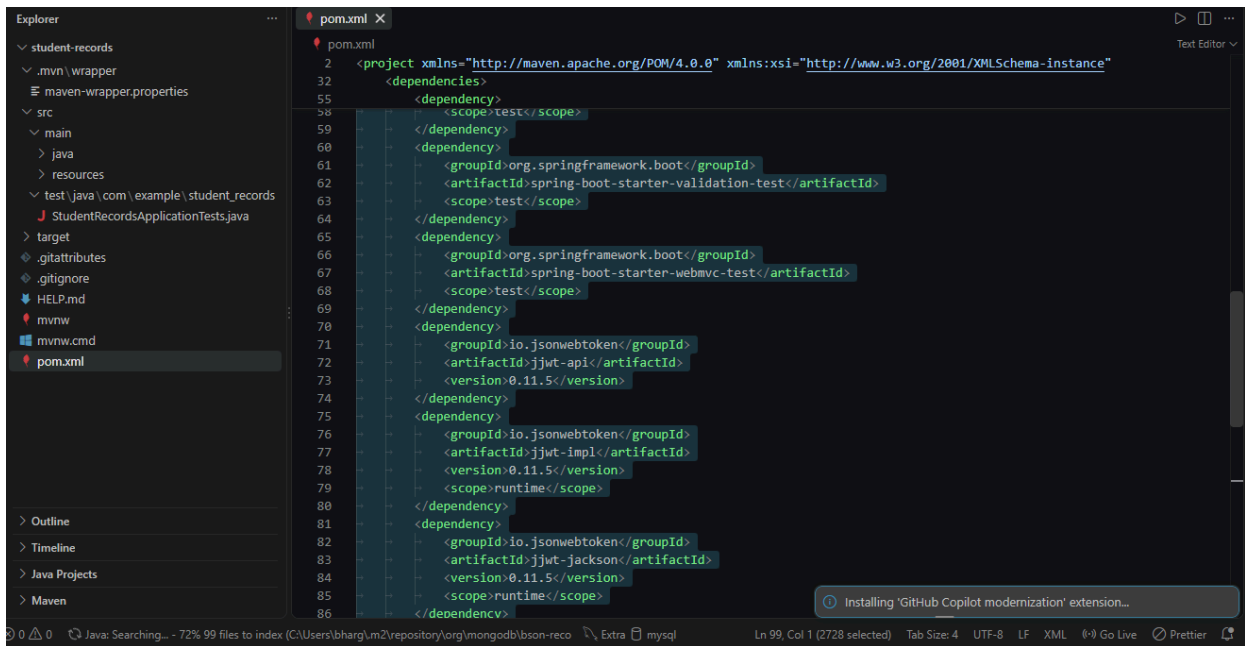
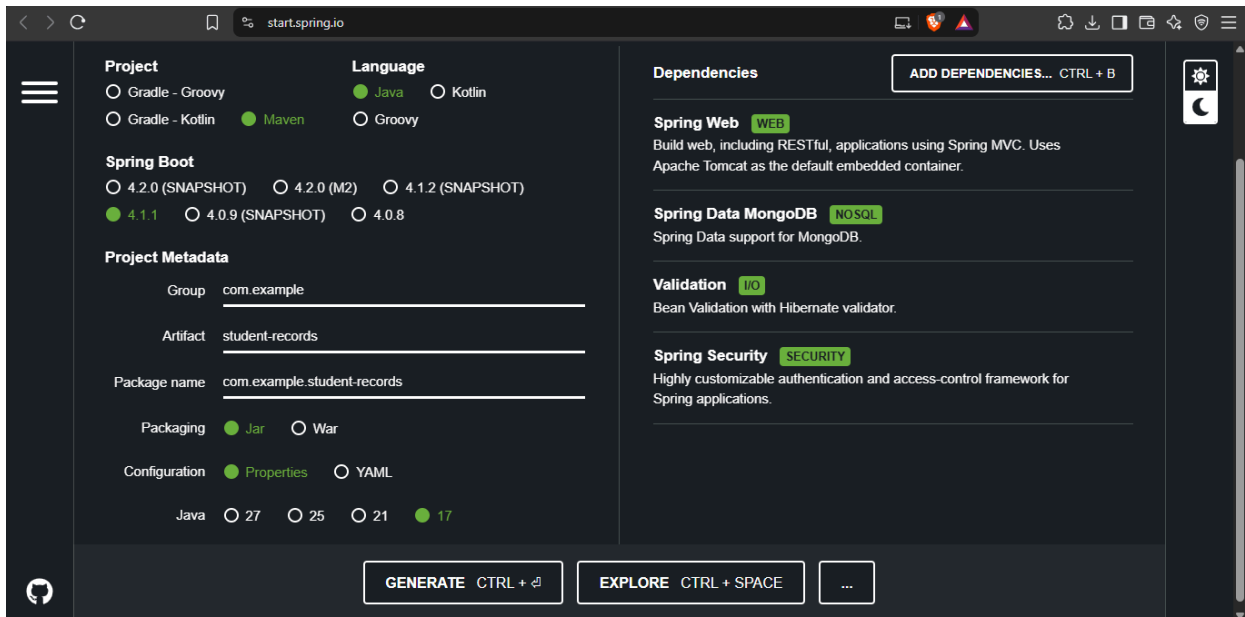
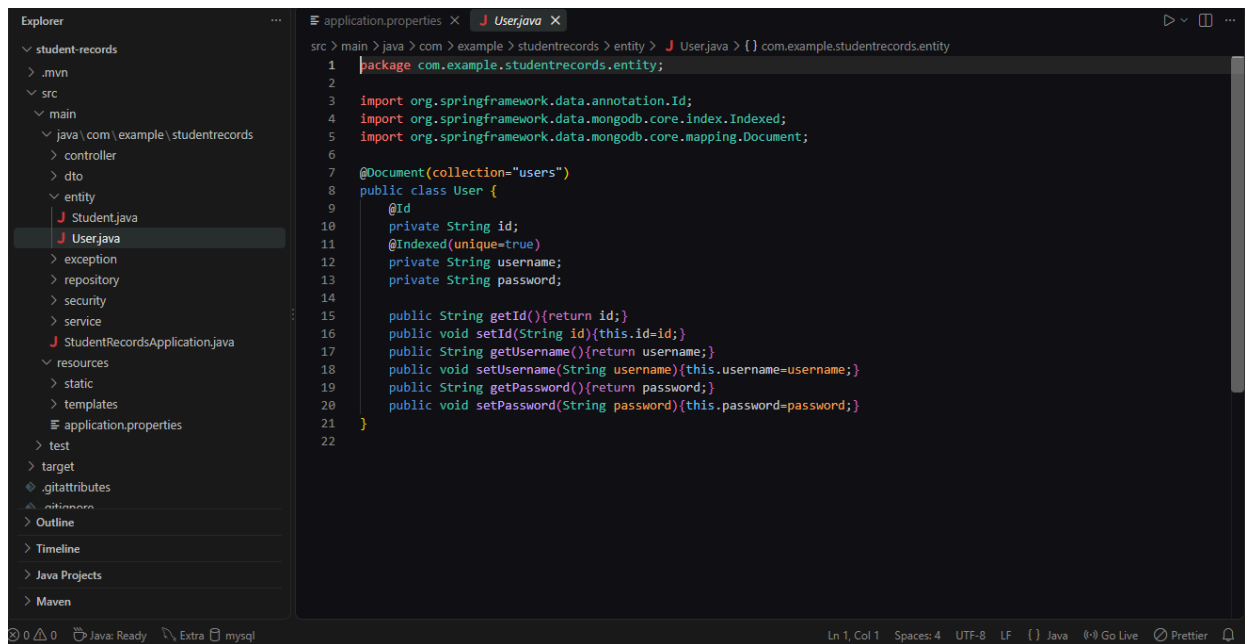
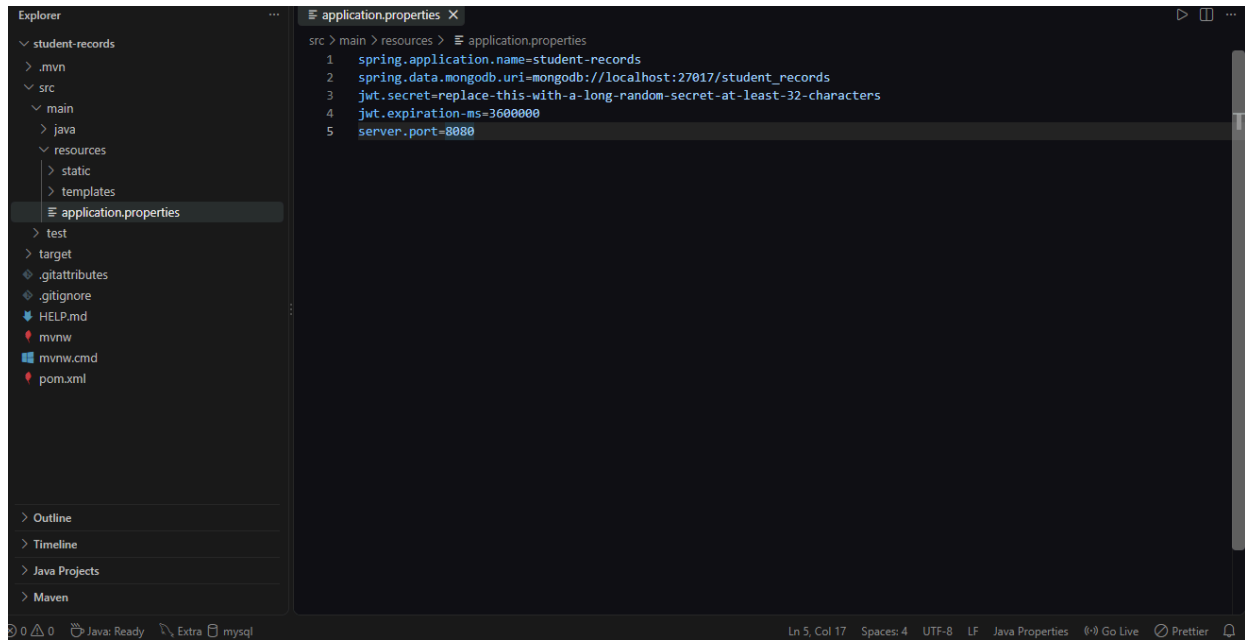
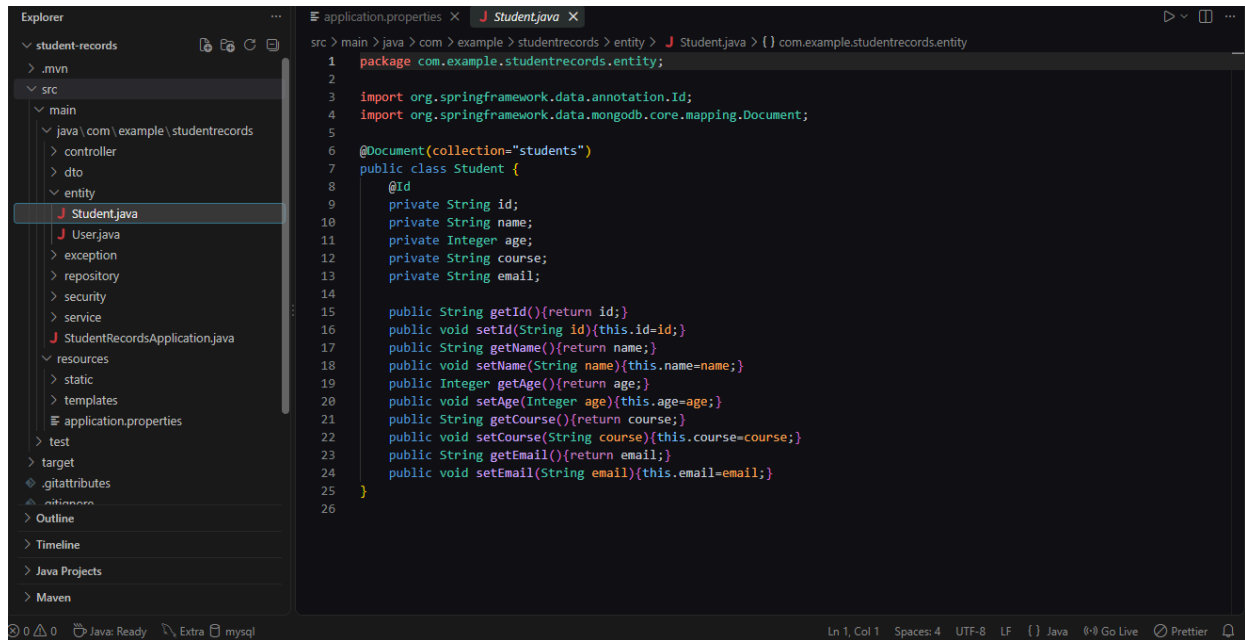
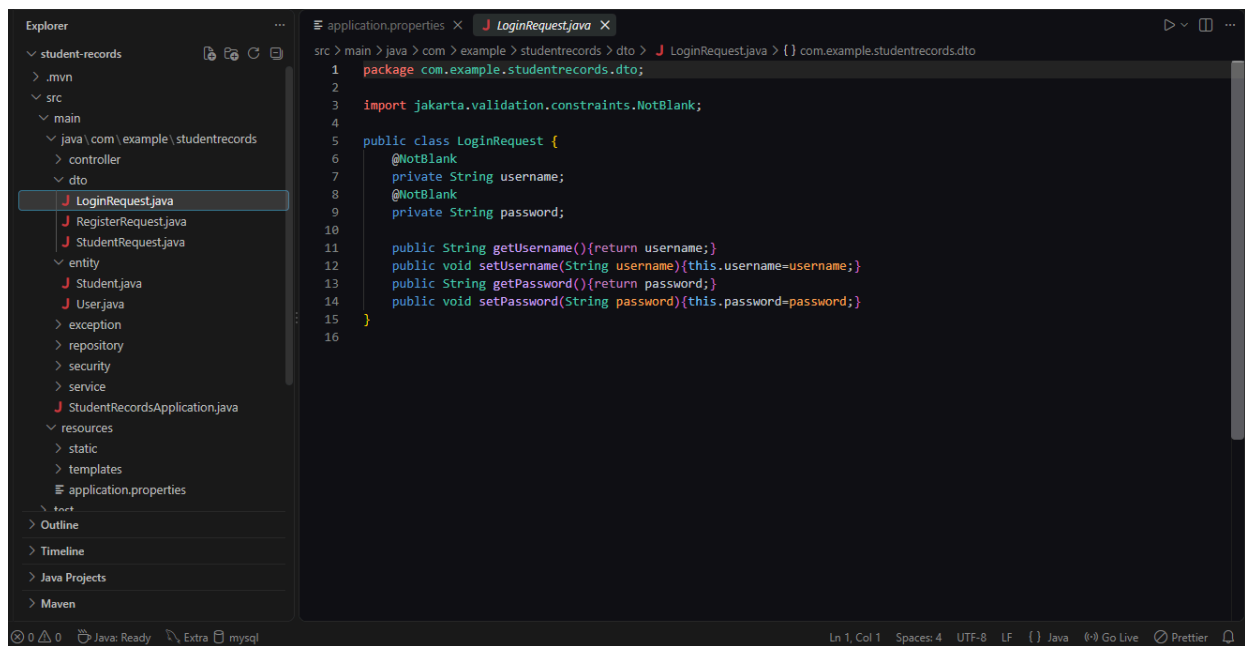




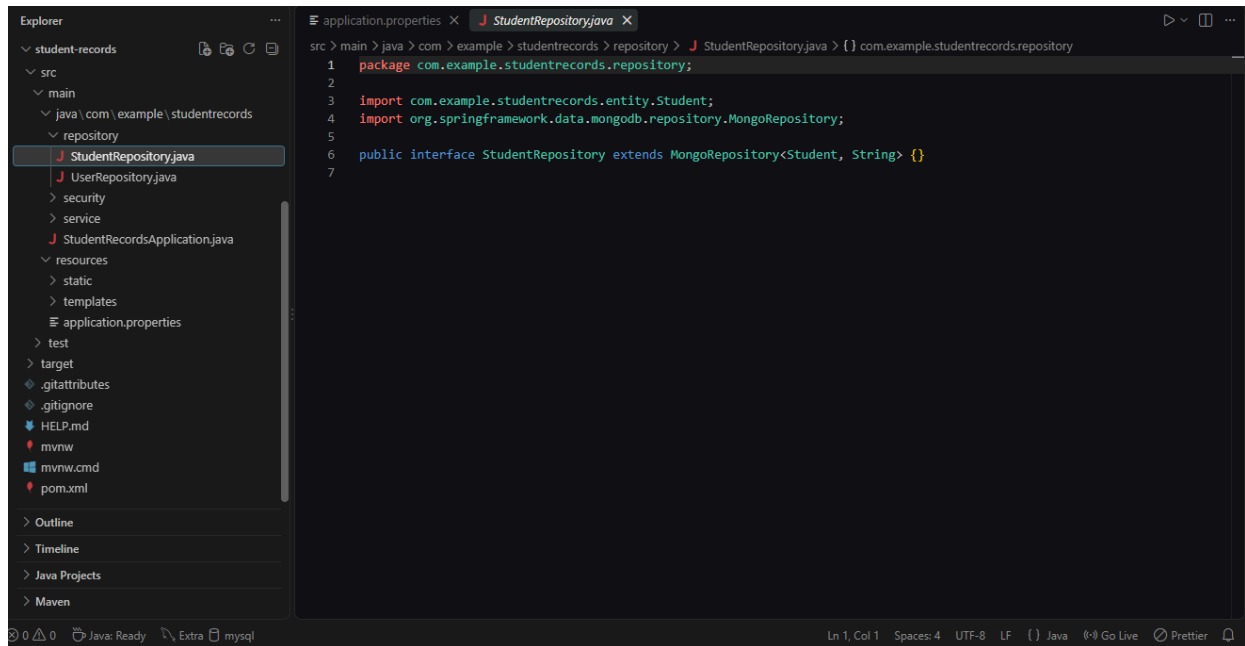




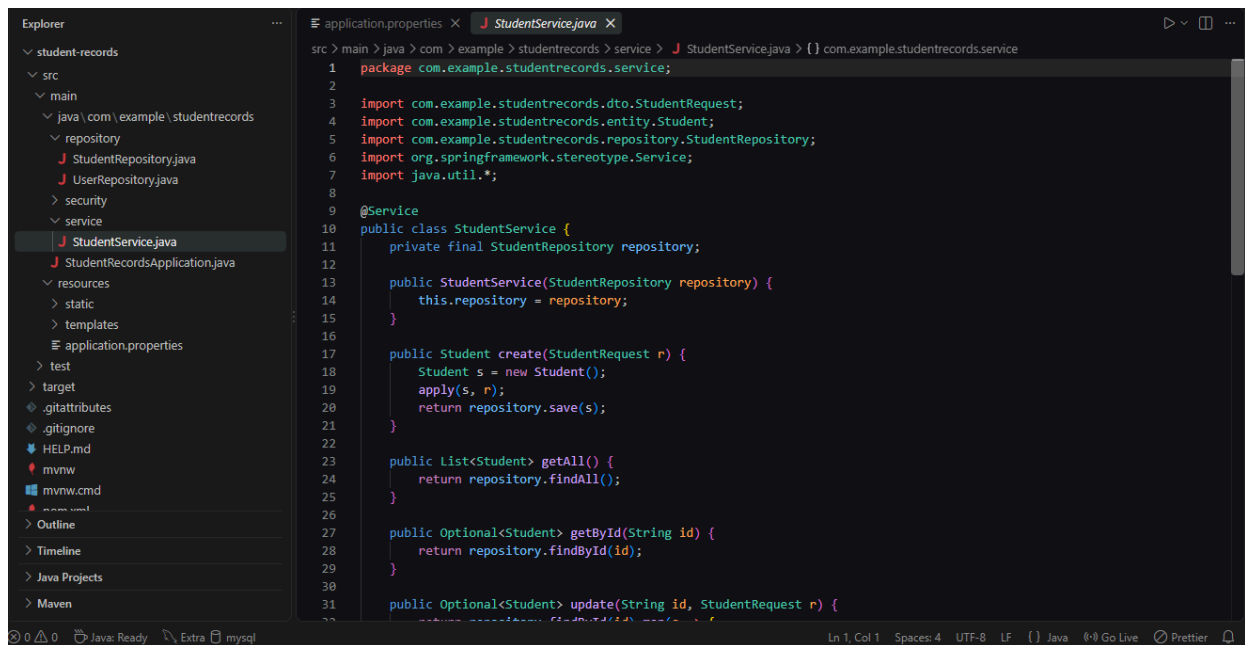
```
src > main > java > com > example > studentrecords > entity > Student.java > {} com.example.studentrecords.entity
1 package com.example.studentrecords.entity;
2
3 import org.springframework.data.annotation.Id;
4 import org.springframework.data.mongodb.core.mapping.Document;
5
6 @Document(collection="students")
7 public class Student {
8     @Id
9     private String id;
10    private String name;
11    private Integer age;
12    private String course;
13    private String email;
14
15    public String getId(){return id;}
16    public void setId(String id){this.id=id;}
17    public String getName(){return name;}
18    public void setName(String name){this.name=name;}
19    public Integer getAge(){return age;}
20    public void setAge(Integer age){this.age=age;}
21    public String getCourse(){return course;}
22    public void setCourse(String course){this.course=course;}
23    public String getEmail(){return email;}
24    public void setEmail(String email){this.email=email;}
25 }
26
```



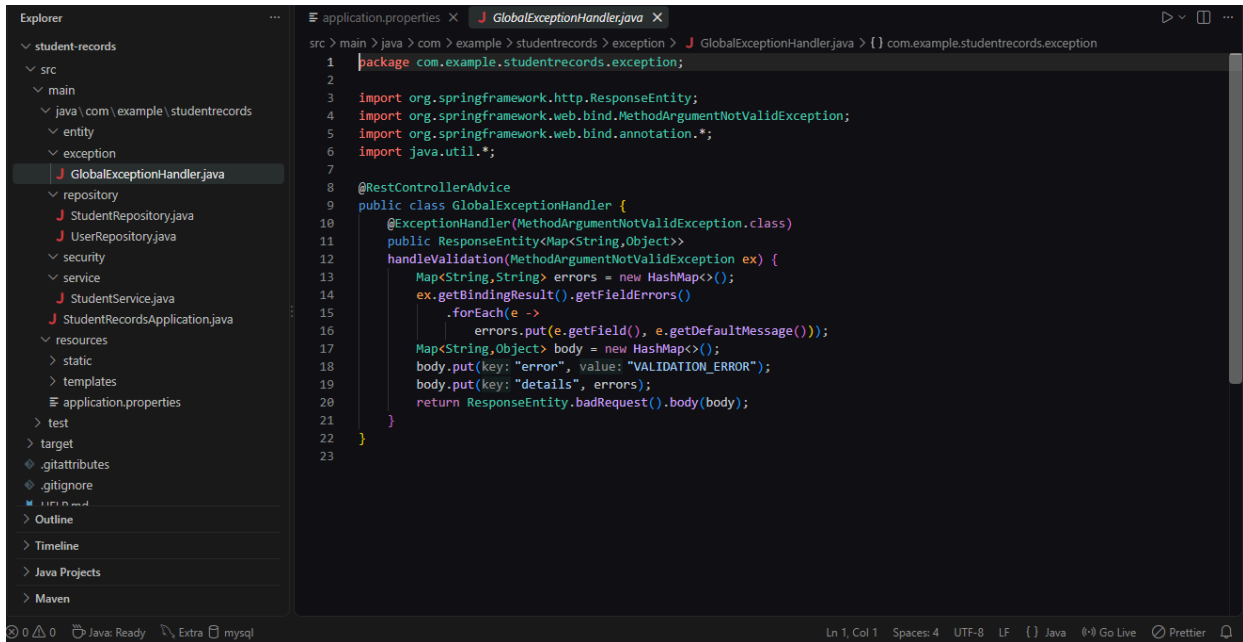
```
src > main > java > com > example > studentrecords > dto > LoginRequest.java > {} com.example.studentrecords.dto
1 package com.example.studentrecords.dto;
2
3 import jakarta.validation.constraints.NotBlank;
4
5 public class LoginRequest {
6     @NotBlank
7     private String username;
8     @NotBlank
9     private String password;
10
11    public String getUsername(){return username;}
12    public void setUsername(String username){this.username=username;}
13    public String getPassword(){return password;}
14    public void setPassword(String password){this.password=password;}
15 }
16
```



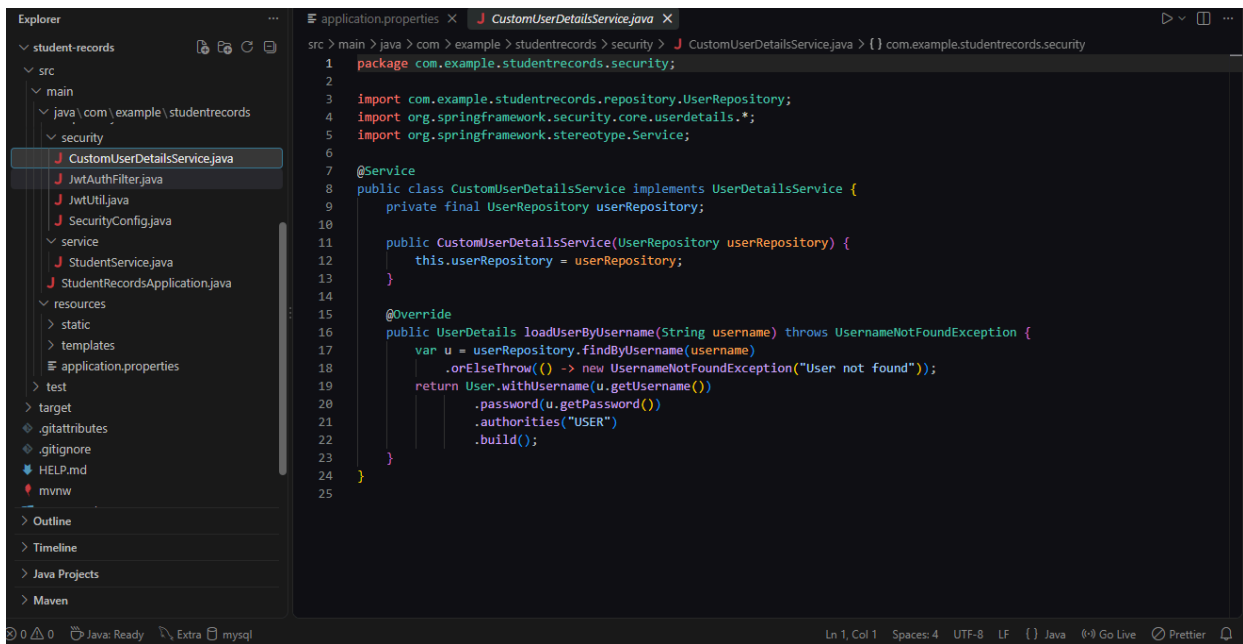
```
src > main > java > com > example > studentrecords > repository > J StudentRepository.java > {} com.example.studentrecords.repository
1 package com.example.studentrecords.repository;
2
3 import com.example.studentrecords.entity.Student;
4 import org.springframework.data.mongodb.repository.MongoRepository;
5
6 public interface StudentRepository extends MongoRepository<Student, String> {}
7
```



```
src > main > java > com > example > studentrecords > service > J StudentService.java > {} com.example.studentrecords.service
1 package com.example.studentrecords.service;
2
3 import com.example.studentrecords.dto.StudentRequest;
4 import com.example.studentrecords.entity.Student;
5 import com.example.studentrecords.repository.StudentRepository;
6 import org.springframework.stereotype.Service;
7 import java.util.*;
8
9 @Service
10 public class StudentService {
11     private final StudentRepository repository;
12
13     public StudentService(StudentRepository repository) {
14         this.repository = repository;
15     }
16
17     public Student create(StudentRequest r) {
18         Student s = new Student();
19         apply(s, r);
20         return repository.save(s);
21     }
22
23     public List<Student> getAll() {
24         return repository.findAll();
25     }
26
27     public Optional<Student> getById(String id) {
28         return repository.findById(id);
29     }
30
31     public Optional<Student> update(String id, StudentRequest r) {
32         return repository.findById(id).map(s -> {
33             s = apply(s, r);
34             return repository.save(s);
35         });
36     }
37
38     private Student apply(Student s, StudentRequest r) {
39         s.setName(r.getName());
40         s.setEmail(r.getEmail());
41         return s;
42     }
43 }
```



```
src > main > java > com > example > studentrecords > exception > GlobalExceptionHandler.java > {} com.example.studentrecords.exception
1 package com.example.studentrecords.exception;
2
3 import org.springframework.http.ResponseEntity;
4 import org.springframework.web.bind.MethodArgumentNotValidException;
5 import org.springframework.web.bind.annotation.*;
6 import java.util.*;
7
8 @RestControllerAdvice
9 public class GlobalExceptionHandler {
10     @ExceptionHandler(MethodArgumentNotValidException.class)
11     public ResponseEntity<Map<String, Object>>
12     handleValidation(MethodArgumentNotValidException ex) {
13         Map<String, String> errors = new HashMap<>();
14         ex.getBindingResult().getFieldErrors().
15             .forEach(e ->
16                 errors.put(e.getField(), e.getDefaultMessage()));
17         Map<String, Object> body = new HashMap<>();
18         body.put(key: "error", value: "VALIDATION_ERROR");
19         body.put(key: "details", errors);
20         return ResponseEntity.badRequest().body(body);
21     }
22 }
23
```



```
src > main > java > com > example > studentrecords > security > CustomUserDetailsService.java > {} com.example.studentrecords.security
1 package com.example.studentrecords.security;
2
3 import com.example.studentrecords.repository.UserRepository;
4 import org.springframework.security.core.userdetails.*;
5 import org.springframework.stereotype.Service;
6
7 @Service
8 public class CustomUserDetailsService implements UserDetailsService {
9     private final UserRepository userRepository;
10
11     public CustomUserDetailsService(UserRepository userRepository) {
12         this.userRepository = userRepository;
13     }
14
15     @Override
16     public UserDetails loadUserByUsername(String username) throws UsernameNotFoundException {
17         var u = userRepository.findByUsername(username)
18             .orElseThrow(() -> new UsernameNotFoundException("User not found"));
19         return User.withUsername(u.getUsername())
20             .password(u.getPassword())
21             .authorities("USER")
22             .build();
23     }
24 }
25
```



```
Problems Output Debug Console Ports
Terminal
PS C:\Users\bhang\Downloads\TriSem\SEM 4\DBMS\Week 8\student-records> Invoke-RestMethod -Uri "http://localhost:8080/auth/register" -Method Post -ContentType "application/json" -Body '{"username":"asha","password":"secret123"}'

message
-----
registered successfully

PS C:\Users\bhang\Downloads\TriSem\SEM 4\DBMS\Week 8\student-records>
```

```
Problems Output Debug Console Ports
Terminal
PS C:\Users\bhang\Downloads\TriSem\SEM 4\DBMS\Week 8\student-records> Invoke-RestMethod -Uri "http://localhost:8080/auth/register" -Method Post -ContentType "application/json" -Body '{"username":"asha","password":"secret123"}'

PS C:\Users\bhang\Downloads\TriSem\SEM 4\DBMS\Week 8\student-records> $token = (Invoke-RestMethod -Uri "http://localhost:8080/auth/login" -Method Post -ContentType "application/json" -Body '{"username":"asha","password":"secret123"}').token
>> $token
eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJhc2hhIiwiaWF0IjoxNzkwNjYxMjExLjE4IiwiaWF0IjE3OTA2NjQ4MTF9.SMgTGfD8z8afVW7nke1P_7c2CvnnvQBQmzs3m9RHrBY0
PS C:\Users\bhang\Downloads\TriSem\SEM 4\DBMS\Week 8\student-records> Invoke-RestMethod -Uri "http://localhost:8080/students" -Method Post -Headers @{ Authorization = "Bearer $token" } -ContentType "application/json" -Body '{"name":"Asha Rao","age":20,"course":"B.Tech CSE","email":"asha@example.com"}'

age : 20
course : B.Tech CSE
email : asha@example.com
id : 6abb5274ec3220c0dac6ebf4
name : Asha Rao

PS C:\Users\bhang\Downloads\TriSem\SEM 4\DBMS\Week 8\student-records>
```

```
Terminal
PS C:\Users\bhang\Downloads\TriSem\SEM 4\DBMS\Week 8\student-records> Invoke-RestMethod -Uri "http://localhost:8080/students" -Method Post -Headers @{ Authorization = "Bearer $token" } -ContentType "application/json" -Body '{"name":"Asha Rao","age":20,"course":"B.Tech CSE","email":"asha@example.com"}'

id : 6abb5274ec3220c0dac6ebf4
name : Asha Rao

PS C:\Users\bhang\Downloads\TriSem\SEM 4\DBMS\Week 8\student-records> Invoke-RestMethod -Uri "http://localhost:8080/students" -Method Get -Headers @{ Authorization = "Bearer $token" }

age : 20
course : B.Tech CSE
email : asha@example.com
id : 6abb5274ec3220c0dac6ebf4
name : Asha Rao

PS C:\Users\bhang\Downloads\TriSem\SEM 4\DBMS\Week 8\student-records>
```

```
Terminal powershell + ~ [ ] [ ] ... [ ] [ ] [ ]
PS C:\Users\bharg\Downloads\TriSem\SEM 4\DBMS\Week 8\student-records> try {
>>   Invoke-RestMethod -Uri "http://localhost:8080/students" -Method Post -Headers @{ Authorization = "Bearer $token" } -Content
tType "application/json" -Body '{"name":"","age":10,"course":"","email":"not-an-email"}'
>> } catch {
>>   $_.Exception.Response.GetResponseStream() | ForEach-Object { (New-Object System.IO.StreamReader($_)).ReadToEnd() }
>> }
{"details":{"name":"name is required","course":"course is required","email":"a valid email is required","age":"age must be at lea
st 16"},"error":"VALIDATION_ERROR"}
PS C:\Users\bharg\Downloads\TriSem\SEM 4\DBMS\Week 8\student-records>
```

Ln 1, Col 1 Spaces: 4 UTF-8 LF { } Java (Go Live) Prettier

Service

- student_recordss 8.3.7
 - admin
 - config
 - local
 - studentDB
 - test 80K
 - Query
 - Collections (2)
 - students
 - Columns
 - _id Object ID
 - name String
 - age Number
 - course String
 - email String
 - _class String
 - Index
 - users

students

Properties DATA {} test

```
db('test').collection('students').find({}).limit(100).toArray()
```

Search Results

	_id	name	age	course	email	_class
	Object ID	String	Number	String	String	String
	ObjectID('6abb5274ec3220k	Asha Rao	20	B.Tech CSE	asha@example.com	com.example.studentrecord

_id ObjectID('6abb5274ec3220c0dac6ebf4')

name Asha Rao

age 20

course B.Tech CSE

email asha@example.com

_class com.example.studentrecords.entity.Student

Java: Ready student_recordss test

Service

- student_recordss 8...
 - admin
 - config
 - local
 - studentDB
 - test 80K
 - Query
 - Collections (2)
 - students
 - Columns
 - _id Object ID
 - name String
 - age Number
 - course String
 - email String
 - _class String
 - Index
 - users
 - Columns
 - _id Object ID
 - username String
 - password String
 - _class String
 - Index

users

Properties DATA {} test

```
db('test').collection('users').find({}).limit(100).toArray()
```

Search Results

	_id	username	password	_class
	Object ID	String	String	String
	ObjectID('6abb5221ec3220k	asha	\$2a\$10\$bncAP0MuGeeC4K	com.example.studentrecord

_id ObjectID('6abb5221ec3220c0dac6ebf3')

username asha

password \$2a\$10\$bncAP0MuGeeC4KijJynUuOso/Q8Teu5GLZ0KeH4QC6XjukRsQkGG2

_class com.example.studentrecords.entity.User

name Nullable

Java: Ready student_recordss test